

Design specifications for AR glasses



I. Overview

 **This document specification is mainly used for Optical Waveguide series such as X2**

If you need UI materials mentioned in the document, you can add discord to get them.

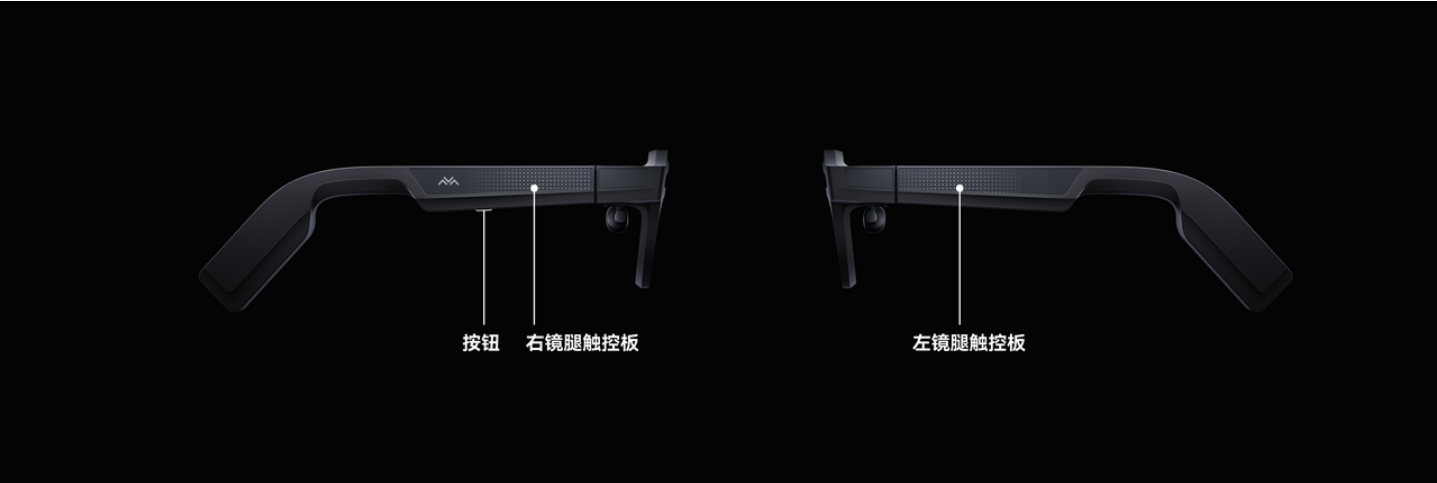
II. Interaction Design Specification

1. Interaction pattern

In RayNeo X2, different ways of interacting with the interface are supported in different scenarios to achieve higher efficiency. The two main interaction patterns are: mirror leg touchpad interaction and ring interaction.

1.1 Mirror leg touchpad interaction

RayNeo X2's temple touch is the most basic interaction pattern. In the temple touch area, there are mainly two types of gestures: **click and swipe** . Developers can design different interaction patterns for the right temple based on these gestures. Currently, it **only supports defining the interaction pattern of the right temple** , and does not support developers defining the interaction of the left temple.



Below we show the basic interaction definition of the right temple touch in the RayNeo X2 system. Developers need to **follow the basic definition of the system** when defining temple interaction.

Interaction region	Interaction pattern	Role
Right temple touchpad	Click	Confirm/Enter
	Double click	Cancel/Return
	Slide from front to back	Move to the previous focus/scroll up the page
	Slide forward from behind	Move to the next focus/Scroll down the page
	Long press	System-level operation, does not support customization

1.2 Ring interaction

Users can interact with RayNeo X2's peripheral device - **Ring** . Using the Ring, users can interact with the system through minimal hand movements such as **sliding/clicking/double-clicking** . RayNeo X2's Ring has two types of Control Units: **Touchpad and Button** , where the Touchpad is the main interaction unit.

Developers **do not need to separately interact with the ring for additional development** , the official APK will automatically map and adapt the interaction operation.



The operation mapping relationship between the ring touchpad and the temple is as follows:

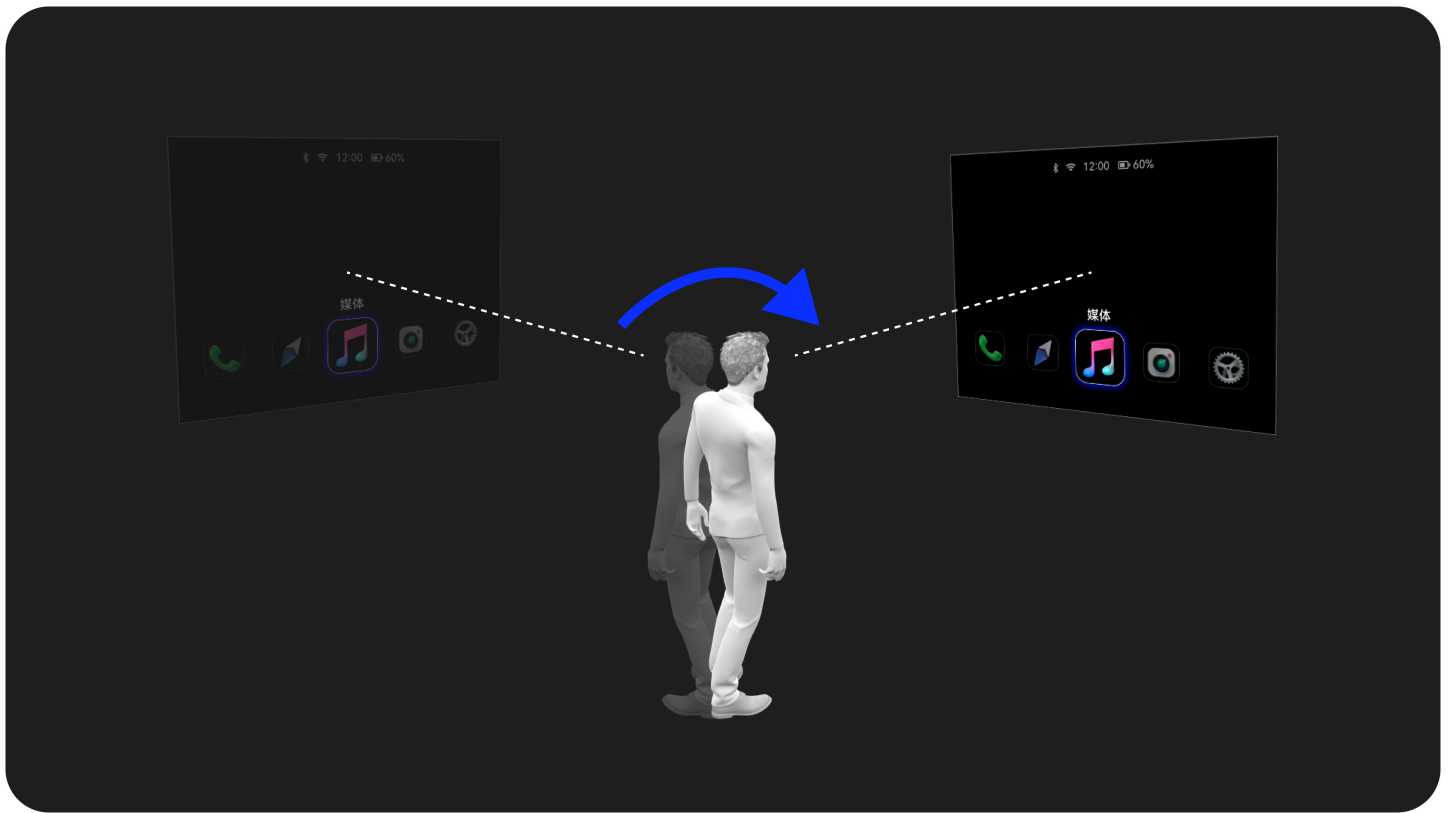
Right temple interaction	Ring trackpad interaction	Role
Click	Click	Confirm/Enter
Double click	Double click	Cancel/Return
Slide from front to back	Swipe from bottom to top/swipe from right to left	Move to the previous focus/scroll up the page
Slide forward from behind	Slide from top to bottom/slide from left to right	Move to the next focus/Scroll down the page
Long press	Long press	System-level operation, does not support customization

2. Display specification

2.1 0, 3, 6 degrees of freedom display effect

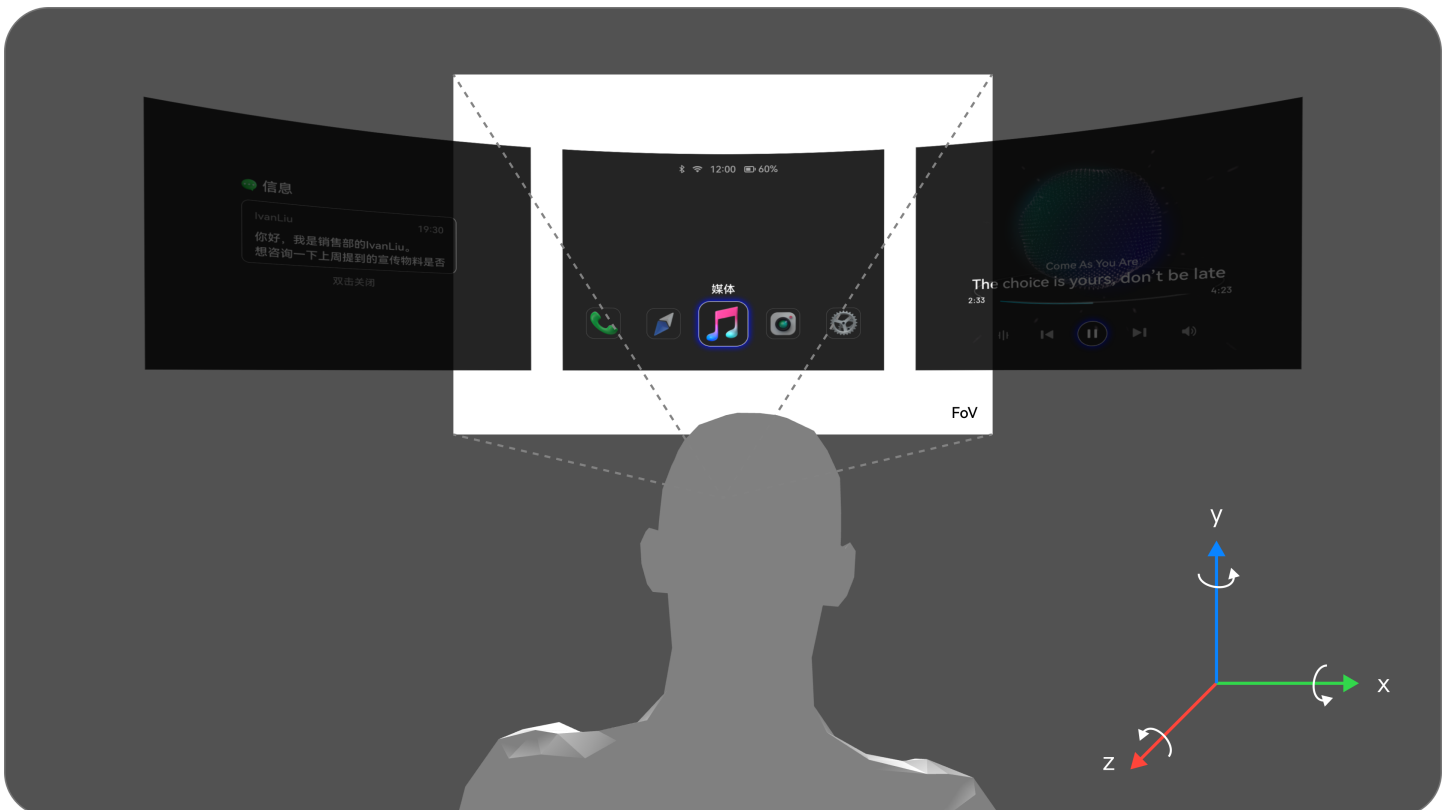
2.1.1 0DOF (0 degrees of freedom)

In a 0-degree-of-freedom scenario, virtual information is fixedly displayed in front of the wearer and moves with the wearer's movement.



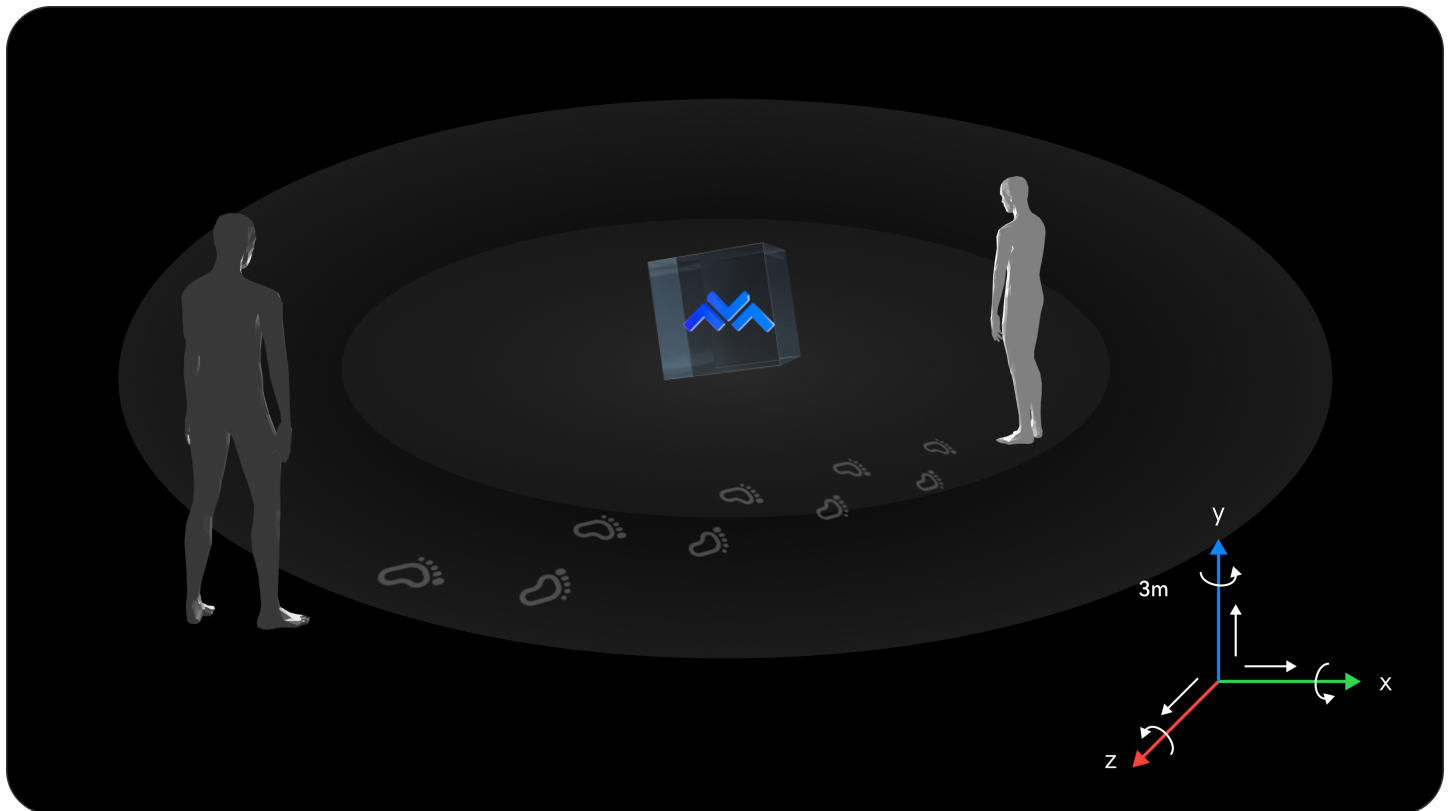
2.1.2 3DOF (3 degrees of freedom)

In a 3-degree-of-freedom scenario, AR glasses can perceive the rotation of the wearer's head. When the wearer rotates their head, they can see different information.



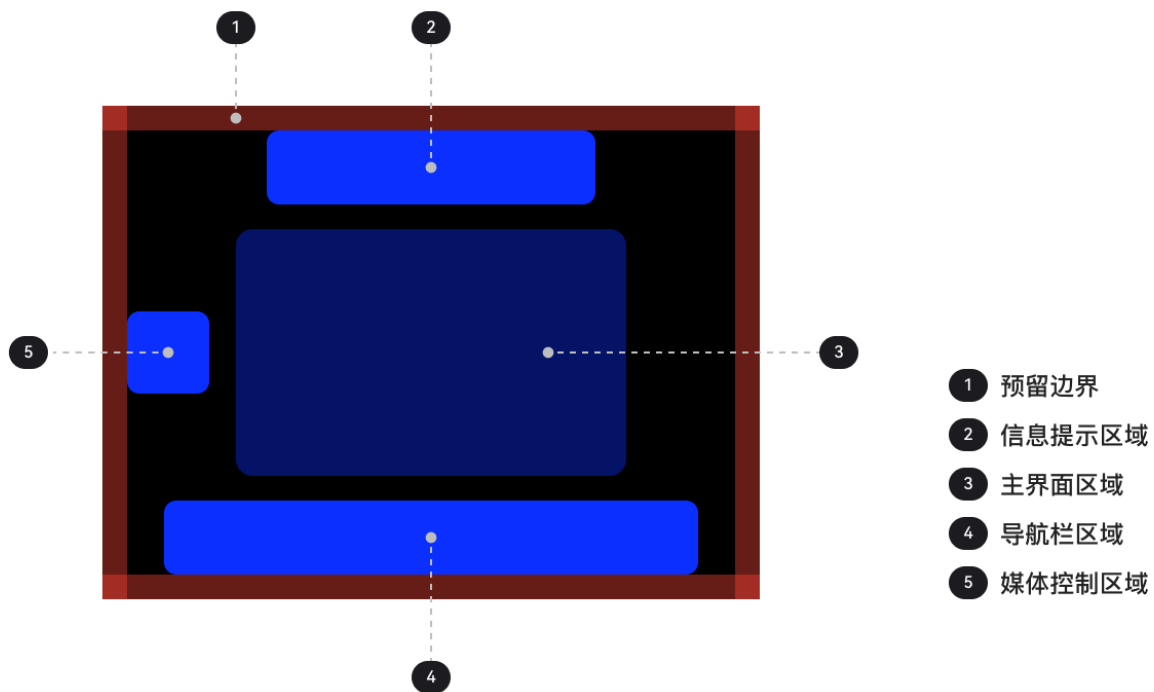
2.1.3 6DOF (6 degrees of freedom)

In a 6-degree-of-freedom scenario, AR glasses can perceive the wearer's head rotation and body displacement. When the wearer rotates their head or moves forward or backward, they can see different information.



2.2 Display area

1. Reserved Boundary
2. Information Prompt Area
3. Main Interface Area
4. Navigation Bar Area
5. Media Control Area



预留边界



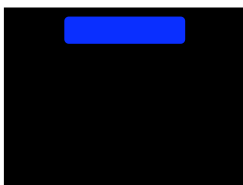
例子



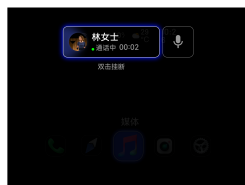
Reserved boundary

A blank area should be reserved around the FoV. At a depth of 5m, 16px boundaries should be reserved around each corner. When the distance is shortened, the reserved boundaries should be appropriately increased; when the distance is widened, the reserved boundaries can be appropriately reduced.

信息提示区域



例子

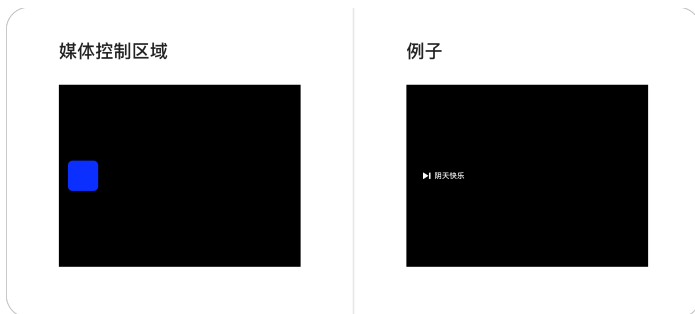
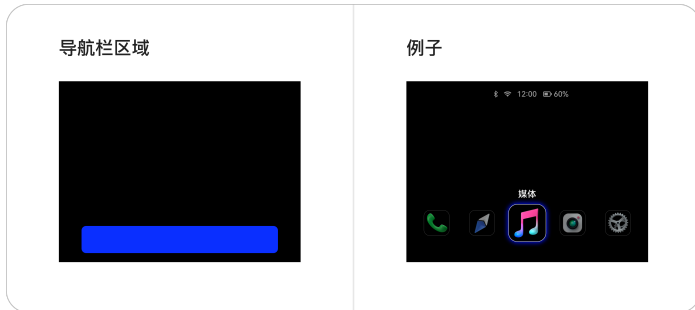
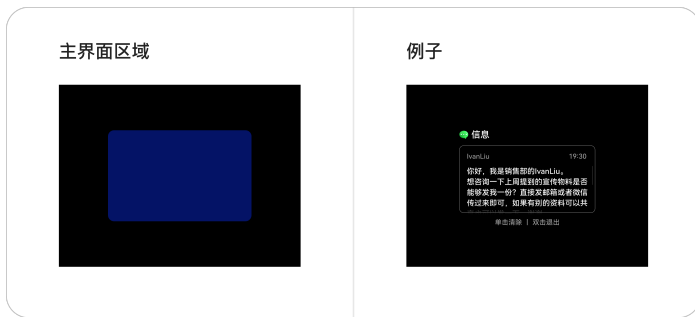


Information prompt area

Low-priority prompt messages that do not require immediate processing by the user can be displayed at the top of the FoV. Such messages disappear after 2-3 seconds of display.

Main interface area

The main interface of the application is displayed in the middle area of the FoV.



Navigation bar area

Navigation bar type information is displayed at the bottom of the FoV.

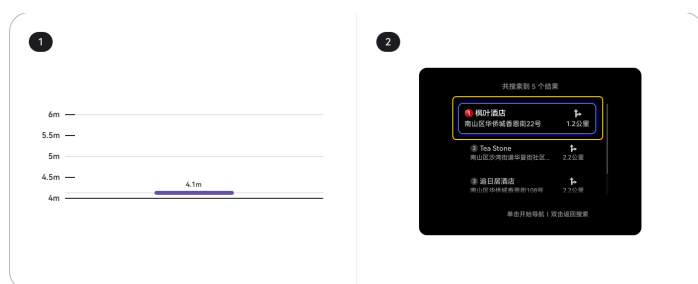
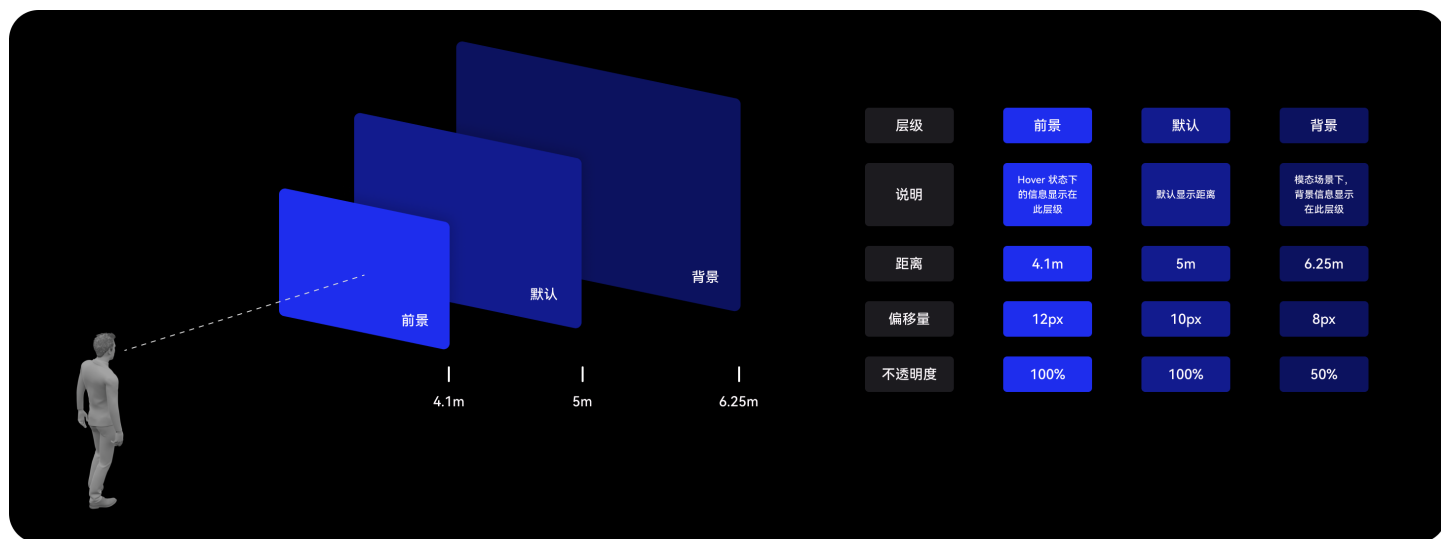
Media control area

Users control media playback through the left mirror leg, so information related to it is displayed on the left side of the FoV to establish a mapping relationship between operation and interface feedback. Such information includes volume adjustment, music playback control, etc.

2.3 Display distance

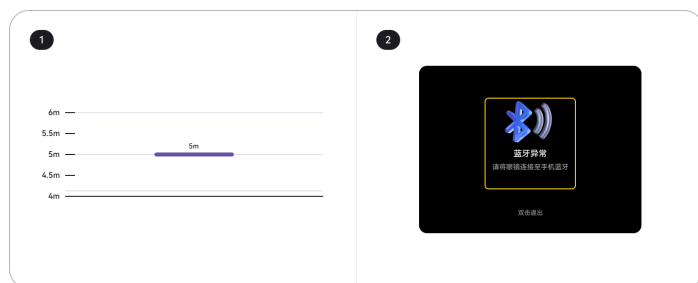
2.3.1 0 dof

In the 0 dof scenario, it is recommended to display veneer information at a distance of 4-6m, as shown in the figure below.



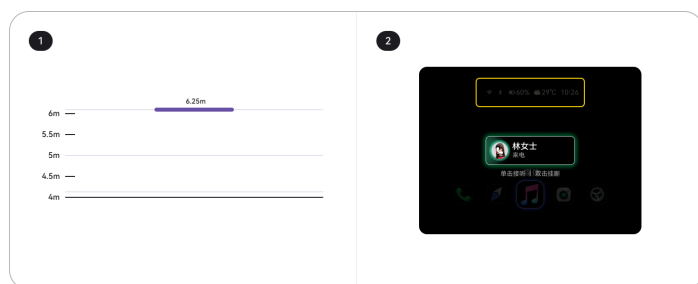
Foreground layer

Hover status information is displayed in the foreground layer at 4.1m. For example, a list in hover status.



Default layer

Non-hover state, readable and interactive information, displaying default layer. For example, exception status prompt.

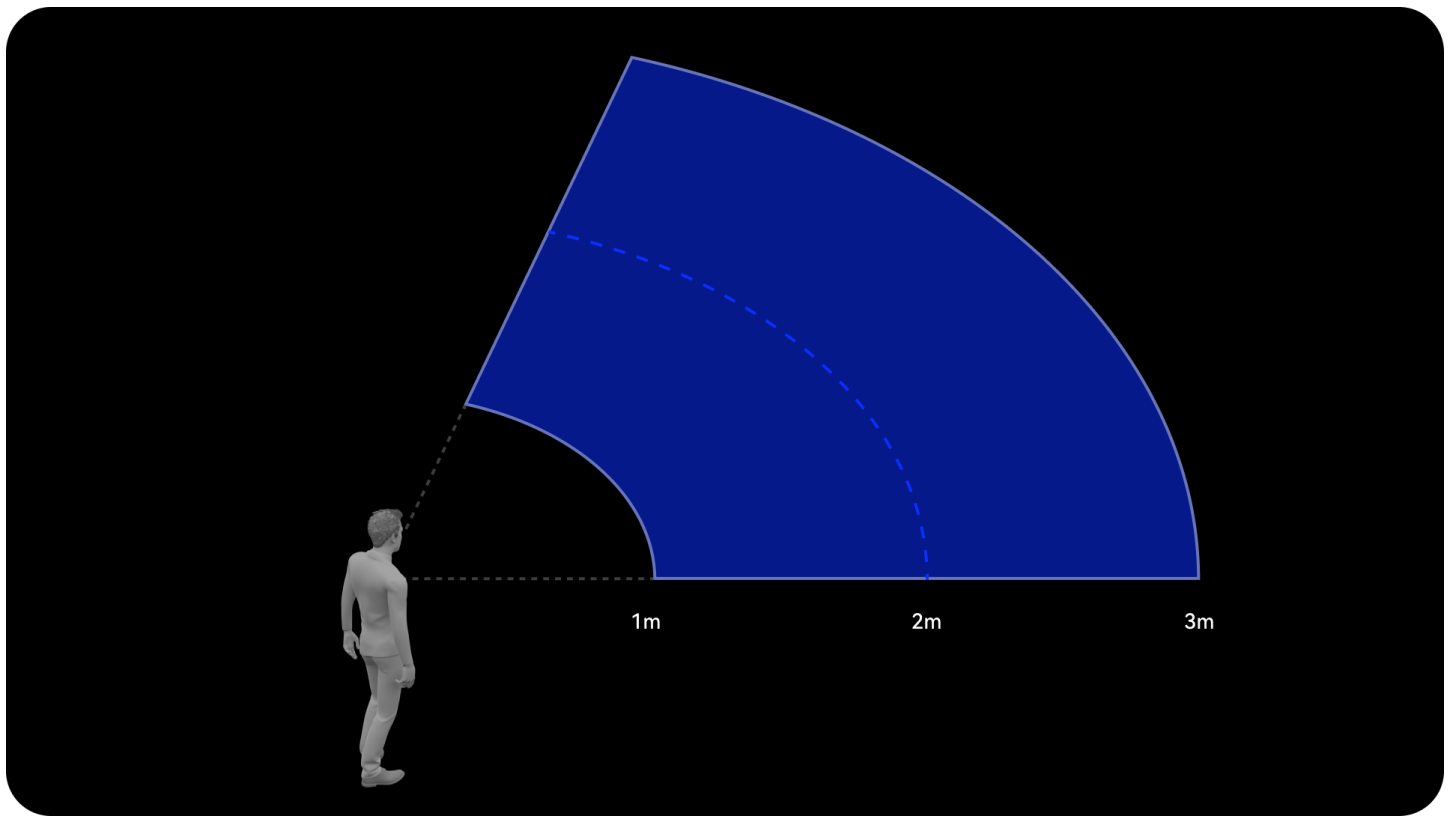


Background layer

When the interface needs to present high-priority information, irrelevant information can be displayed in the background layer and transparency can be increased to help users focus on high-priority information.

2.3.2 3Dof or 6dof

In 3dof or 6dof scenarios, if virtual information can interact with users, it is recommended to display it at a distance of 1-3m.

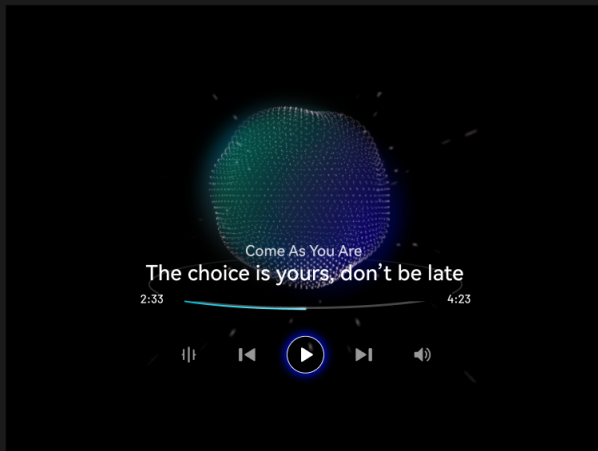


III. Visual Identity Specification

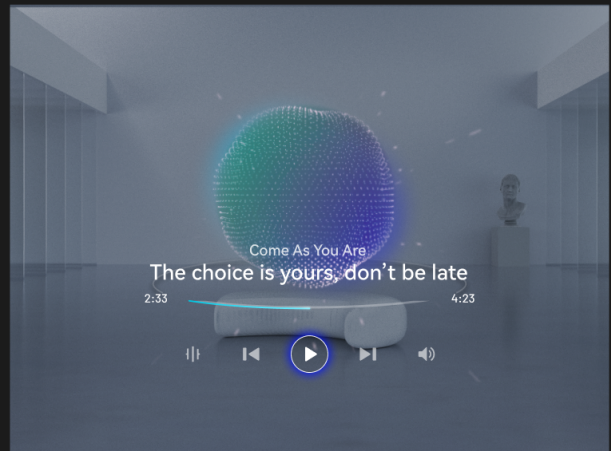
1. AR glasses display features

Light color display feature

In the world of augmented reality (AR), black usually represents areas without light sources, while elements lacking light sources appear transparent to the human eye.



设计界面

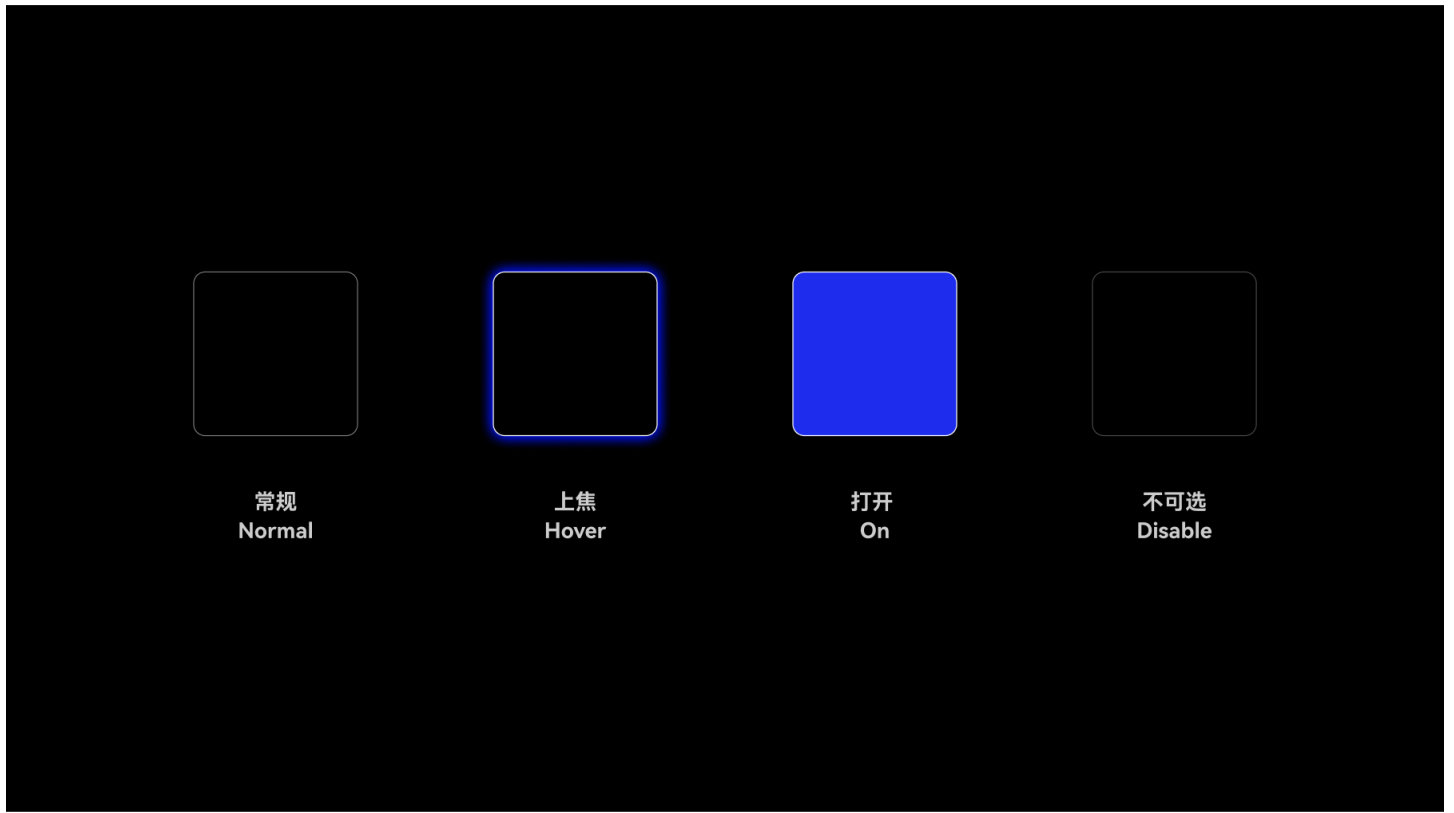


实际看到的界面

2. Visual specification

2.1 Status

During the operation process, the presentation of different information states requires a clear feedback mechanism to ensure that the meanings between states have high recognition and good correlation.



2.2 Icon

Icons need to ensure recognition and help users quickly identify information.

Desktop application icons

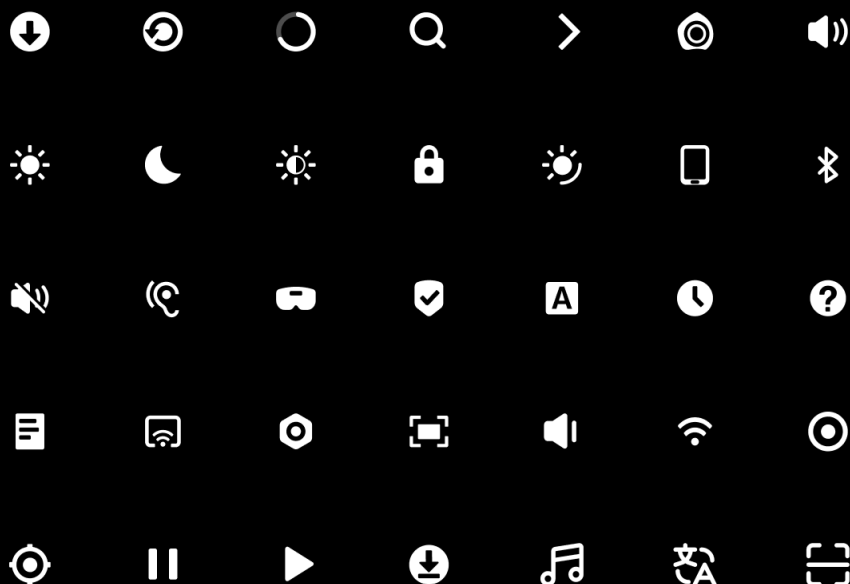
If the application is recommended to use the launcher interface, it is recommended to use 3D style icons.



Function icon

The form of functional icons should be simplified to the most concise elements, without too many details.

The internal elements of the facial icon need to be thicker than the regular icon, because the line in the actual scene will be weakened, which will affect recognition.



2.3 Color

Color selection can affect User Experience and emotional response, and is also used to guide users, establish brand identity, and ensure the accessibility of applications. Therefore, when designing AR applications, it is necessary to be good at selecting and using colors.

Color Usage Guide

Gradual change colors: Use gradual change colors with caution, especially under high contrast conditions, to avoid obvious color scale issues. If necessary, it is recommended to prioritize the gradient effect of the blue channel to minimize the adverse impact of color scale issues on User Experience.

Avoid large areas of use: white, red (only use red for warning colors and reminders).

Use with caution: black, because in the AR world, black represents non-luminous, and for the human eye, non-luminous elements will appear transparent, which may cause the elements to be invisible.

Regular theme color



2.4 Font

Font size

It is recommended to use an appropriate font size to ensure that users can read your text. Please make sure to use an appropriate font size. When developing at a distance of five meters, it is recommended to use the following font size. When the distance (depth of field) is shortened, enlarge the font appropriately, and vice versa.

字号类别	字体 / 字号 / 行高
H1 特殊场景	Medium / 28 / 42
H2 一级标题	Medium / 24 / 32
H3 内容	Medium / 20 / 28
H4 辅助信息	Medium / 18 / 26

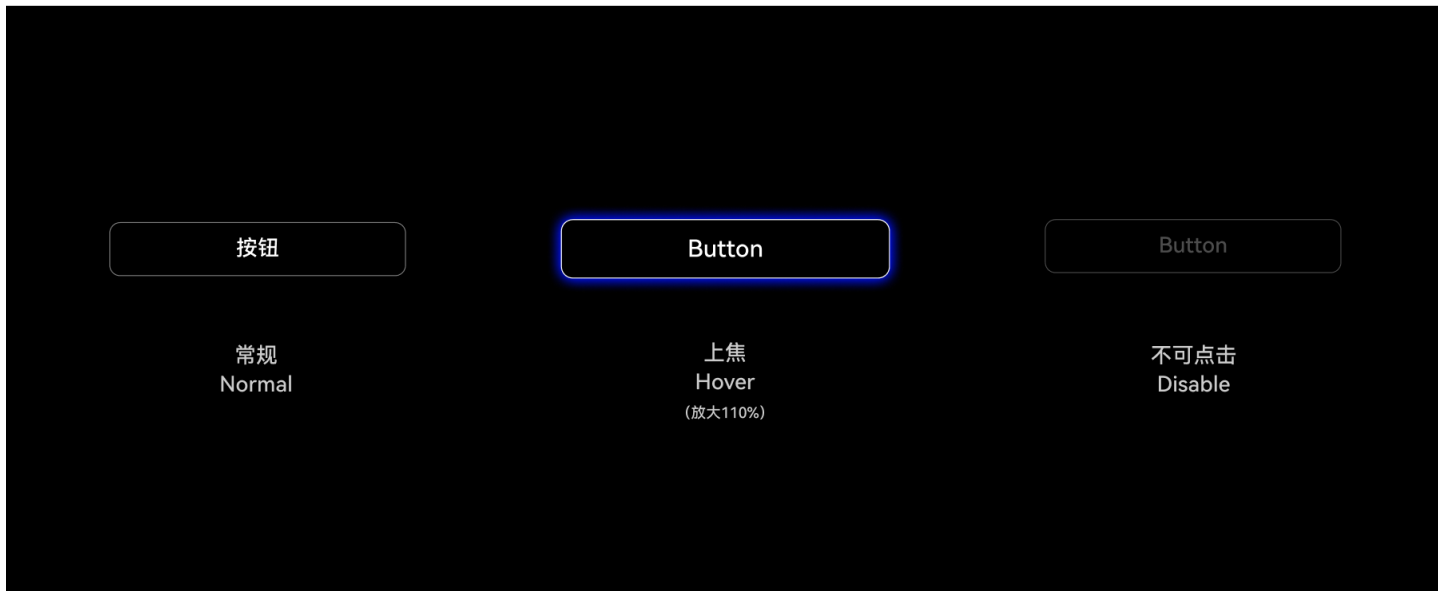
2.5 Spatial hierarchy



3. Controls

1. Button

Button statuses generally include Normal, Hover, and Disable.



Scenario example

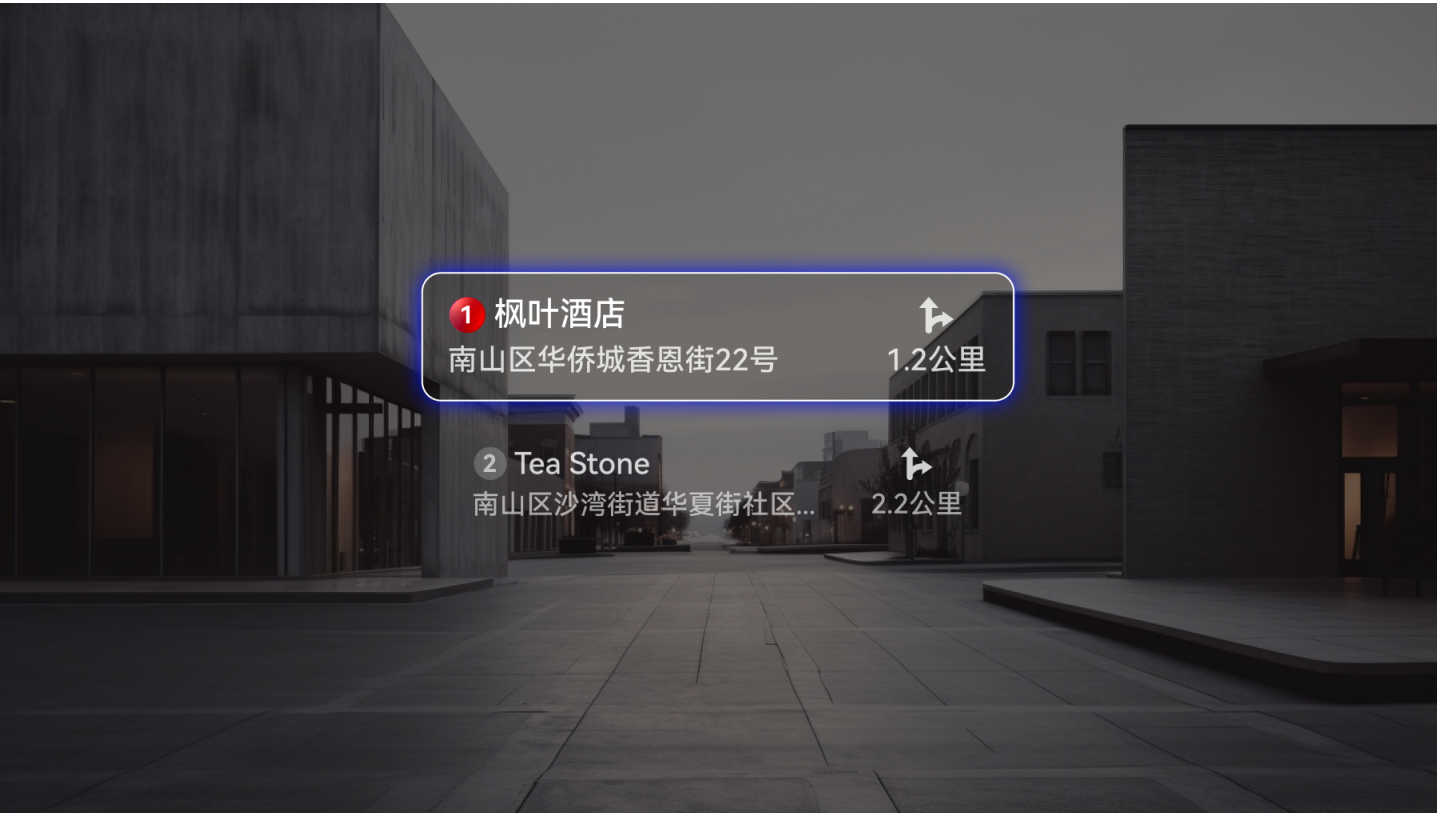


2. List

Lists are divided into single-line lists and double-line lists. Single-line lists display a maximum of one line of text, while double-line lists display only one line of text in the subtitle when unselected, and two lines when selected (hover).



Scenario example



3. Pop-up window

The text is short and clear

The prompt content should be kept short and clear so that it can be read and operated quickly.

Use only when necessary

The use of pop-ups will interrupt the current task process. Please ensure that their timing does not interfere with User Experience and cause trouble.

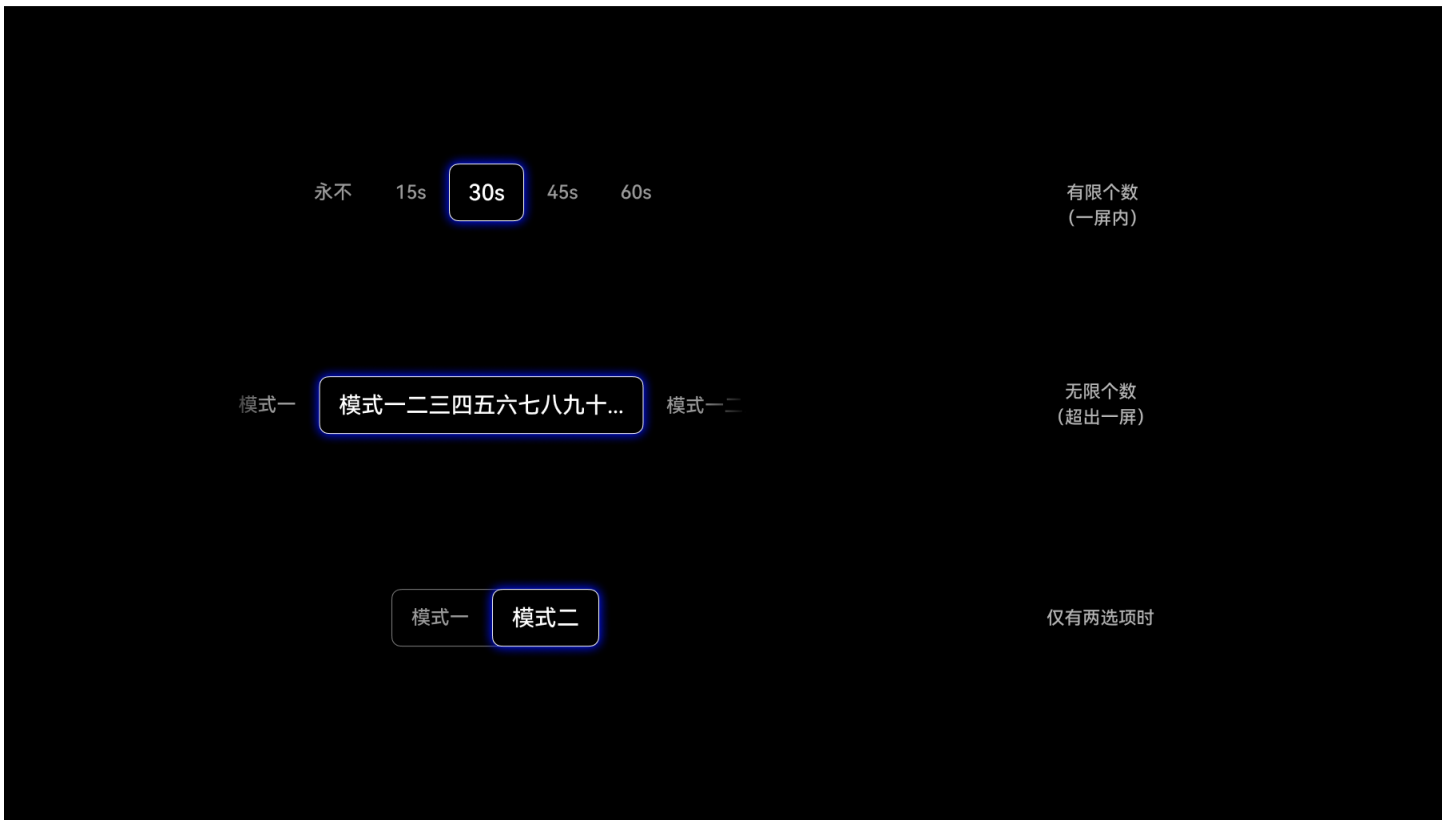


Scenario example



4. Selection bar

Users need to make a single choice from multiple different options.



Scenario example

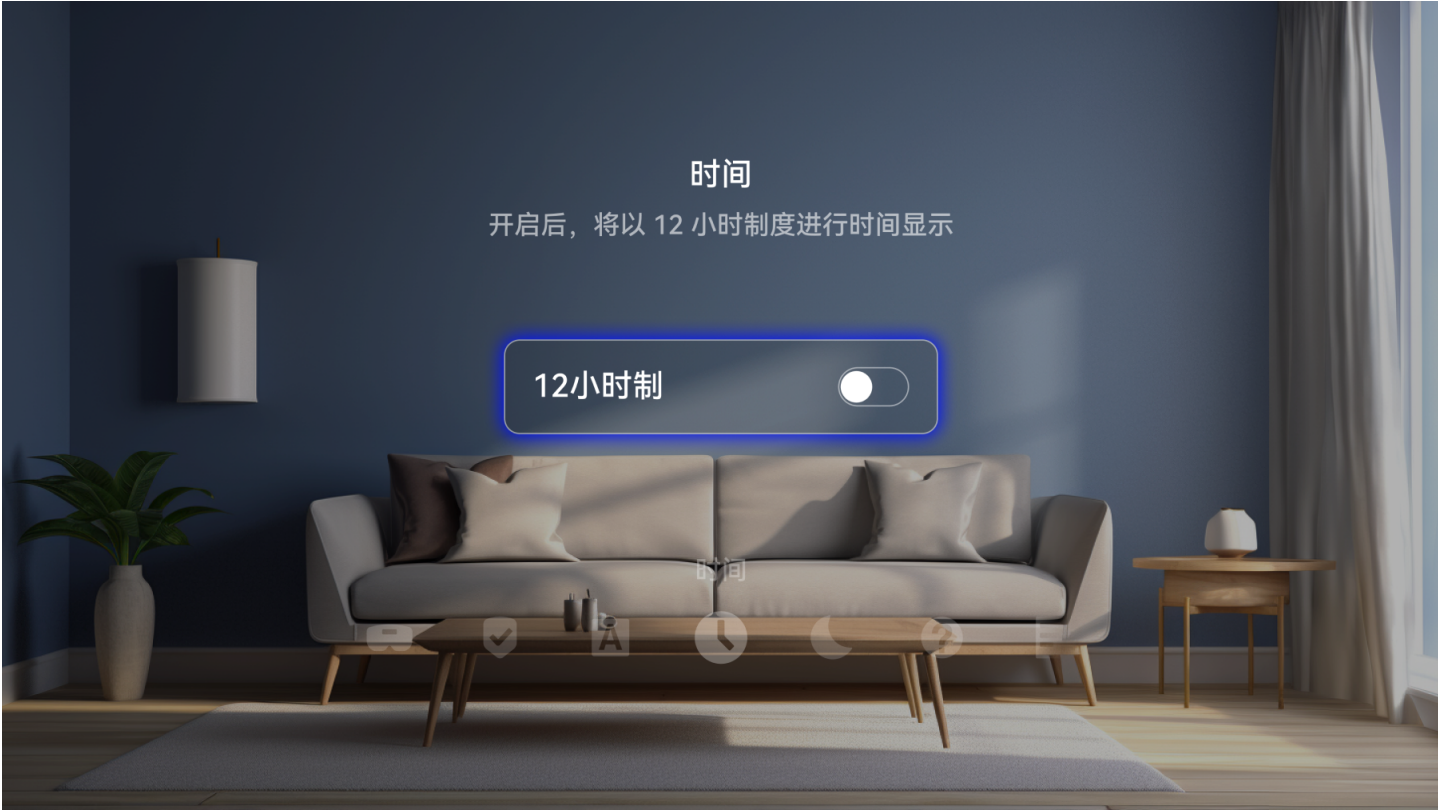


5. Switch

Usually used to enable or disable specific features, options, or settings. Provides users with a simple and intuitive way to control the functions and settings of applications or systems.



Scenario example



6. Progress

Used to display the progress and status of tasks or processes, and to let users know when tasks are completed.



Scene illustration

